Topic4-1

July 18, 2026 12:02 PM

Logistic Regression

Estimation and Interpretations

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In this lecture

- Logistic regression: definition
- Estimation
- Interpretation for different type of covariates
- Inference: confidence intervals and tests

The types of regression models are determined by the types of the responses, not the predictors.

For linear regression models, we assume that the response is continuous and normally distributed.

In practice, many responses are binary:

$$y = \begin{cases} 1 & \text{happy / pass} \ \ldots \\ 0 & \text{unhappy / fail} \ \ldots \end{cases}$$

Logistic Regression

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Logistic regression is a powerful statistical method used for modelling a binary (and binomial) response variable.

For example, we can use a logistic regression to

1. compare the presence of bacteria between groups taking a new drug and a placebo, respectively.
   - Response: *present* or *not present*

2. predict whether or not a customer will default on a loan given their income and demographic variables.
    - Response: default or not default

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3. know how GPA, ACT score, and number of AP classes taken are associated with the probability of getting accepted into a particular university.
    - Response: accepted or not accepted

The response

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To fit a logistic regression in R, the response needs to be a numerical variable (0 and 1) or a `factor`, with two levels (since R stores factors as integers).

For example, a numerical binary response Y_i that flags the successes (S) has the form:

$$Y_i = \begin{cases} 1 \text{ if the } i\text{th observation is S,} \\ 0 \quad \text{otherwise.} \end{cases}$$

In R, you can use the following code to convert a string response into a numerical one:

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```
y_reponse <- if_else(y == "success", 1, 0)
```

Dataset

In this lecture, we'll use the `Titanic` dataset, which contains survival and demographic information about the passangers.

- You can find details about the columns by typing `?titan`.

```
The titanic dataset has information about 891 passengers.
```

passenger_id	survived	passenger_class	name	sex	age	sibsp	parch	ti
759	0 No	3	Theobald, Mr. Thomas Leonard	male	34	0	0	
523	0 No	3	Lahoud, Mr. Sarkis	male	NA	0	0	

Yes

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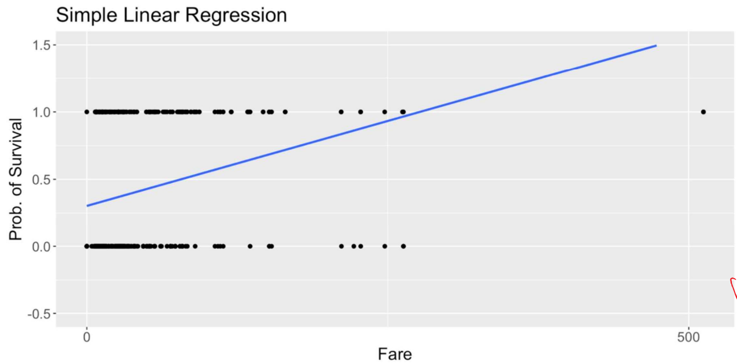
Linear Regression: a range problem

Let's start by using a LR to explore if there is any association between `fare` and `survival`



$$y = \begin{cases} 1 & \text{survived} \\ 0 & \text{dead} \end{cases}$$

$x = \text{fare}$ (confounders)

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A function with range $(0, 1)$

With a LR, fitted values may be larger than 1 or smaller than 0!

> When the response is binary: $E[Y|X] = P[Y|X]$. We are modeling a probability!!

Let's use a *function* of the linear component with range $(0, 1)$:

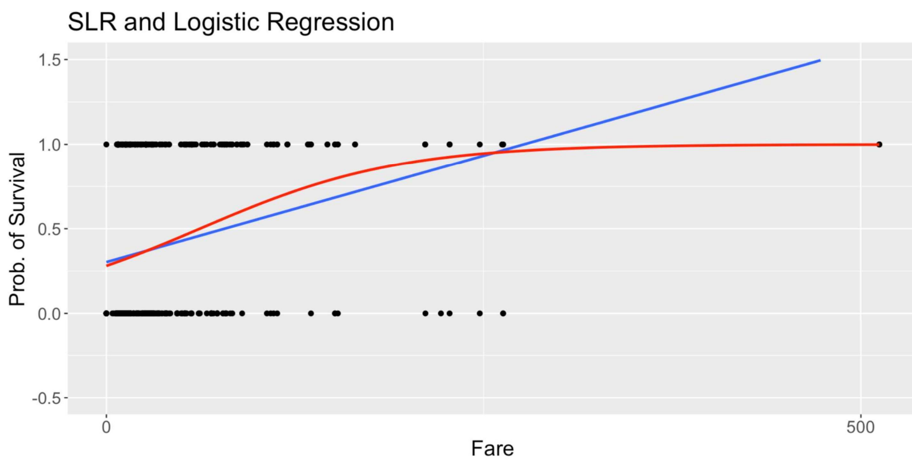
$$E[\text{ survived}_i \mid \text{fare}_i\ ] = P[\text{ survived}_i \mid \text{fare}_i\ ] = \frac{e^{\beta_0+\beta_1\times\text{fare}_i}}{1+e^{\beta_0+\beta_1\times\text{fare}_i}}$$

> Note that this ratio is always between 0 and 1!

$$\log\frac{P(y=1)}{1-P(y=1)} = \beta_0+\beta_1 x$$

$$\Longleftrightarrow P(y=1) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$$

$E(y|x)$


SLR and Logistic Regression
Prob. of Survival
1.5
1.0
0.5
0.0
-0.5
0
500
Fare

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The logit function

Using some algebra you can get an alternative formulation:

$= P(y_i = 1)$

$$P\,[\text{ survived}_i \mid \text{fare}_i\ ] = p_i = \frac{e^{\beta_0+\beta_1\times \text{fare}_i}}{1+e^{\beta_0+\beta_1\times \text{fare}_i}} \qquad \text{[probability]}$$

$i = 1, 2, \ldots, n$

can also be written as

$$\underbrace{\log\left(\frac{p_i}{1-p_i}\right)}_{\textbf{the logit function}} = \beta_0 + \beta_1\,\text{fare}_i \qquad \text{[log-odds]}$$

$\text{logit}(p_i)$

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The Logistic Regression

We model the logit of the probability with a linear component!!

> then all that we learned for MLR can be used for this function of the conditional expectation!!

Thus, this model is known as a **Logistic Regression**

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots + \beta_p X_{pi}$$

linear predictor, denoted by L_i

additive model

> Note that there is no error term! we are modeling a function of the conditional expectation

$p_i = P(y_i = 1) = E(y_i)$

log(odds)

$\log \frac{E(y_i)}{1-E(y_i)} = L_i$

or $\frac{E(y_i)}{1-E(y_i)} = e^{L_i}$ odds

or $E(y_i) = \frac{e^{L_i}}{1+e^{L_i}}$

p_i nonlinear

$e^{\beta_0+\beta_1 X_1} = e^{\beta_0} * e^{\beta_1 X_1}$

linear model $E(y_i) = L_i$

A logistic model can also be written as

$h(E(y_i)) = \beta_0 + \beta_1 X_{1i}$ $\qquad h(x) = \log \frac{x}{1-x}$ logit function

where $h(\cdot)$ is called a link function;

A linear model is

$h(x) = x$, i.e. $E(y_i) = \beta_0 + \beta_1 X_{1i}$

A model in the form of

$\underbrace{h}_{\text{link function}}(E(y_i)) = \underbrace{\beta_0 + \beta_1 X_{1i}}_{\text{linear part}}$ is called a generalized linear model (GLM)

Logistic: a generalized linear model

The logistic regression belongs to a family of models called "generalized linear models", aka GLMs.

In R, we can use the function `glm()` that estimates these models but we need to specify which model in the family we are interested in.

> For logistic regression: `family = binomial`

In general, there is no closed-form to calculate the MLE. Thus, an iterative algorithm is needed (see # iteration is the `summary`)

Bernoulli distribution

Y	0	1
Prob	$1-p$	p

is a special case of Binomial dist (when $n=1$)

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The MLR also belongs to this family and you can also use `glm()` to estimate it. normal distribution

> It is the default family: `family = gaussian` & $h(x) = x \Rightarrow$ linear model

Note that `glm()` calculates maximum likelihood estimators (MLE) and `lm()` calculates least squares estimators.

> For Normal random errors, both methods give the same estimates!

You can get more information about this function using `?glm`

Logit: log-odds

`glm()` models, by default, the logit of the probabilities, usually referred as the "canonical link" function

> links the conditional expectation with a linear component

Note that the **logit of the probabilities** is the log of the odds of success (more details are given in the math appendix)

$$\text{logit}(p) = \log \underbrace{\left(\frac{p}{1-p}\right)}_{\textbf{odds}} = \log(\text{odds})$$

Fitting a Logistic Regression

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Case 1: one continuous covariate

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$$\log\left(\frac{P(\text{survived}_i = 1 \mid \text{fare}_i)}{1 - P(\text{survived}_i = 1 \mid \text{fare}_i)}\right) = \beta_0 + \beta_1 \text{ fare}_i$$

same as liear model

wage2

```r
model_titanic_logistic <- glm(formula = survived ~ fare, data = titan,
                              family = 'binomial')

summary(model_titanic_logistic)
```

```
Call:
glm(formula = survived ~ fare, family = "binomial", data = titan)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.941330   0.095129  -9.895  < 2e-16 ***
fare         0.015197   0.002232   6.810 9.79e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1186.7  on 890  degrees of freedom
Residual deviance: 1117.6  on 889  degrees of freedom
AIC: 1121.6
```

β_0 β_1

p-value very small: survival ~~strongly~~ depends on fair.

$$P(\text{survival}) = \frac{\exp(-0.94 + 0.015\,\text{fare})}{1 + \exp(-0.94 + 0.015\,\text{fare})}$$

assume $\lambda = 1$ may not hold

Interpretation: log-odds

The coefficients can be interpreted in terms of the log-odds (logit link):

- **Intercept** $\hat{\beta}_0 = -0.9413$: if the `fare` paid was 0 dollars, the log-odds of surviving are expected to be -0.9413.

- **Slope** $\hat{\beta}_1 = 0.0152$: an increase of 1 dollar in the `fare` paid is associated with an increase in the log-odds of surviving of 0.0152.

> same as for MLR but with response "log-odds of success"!

Interpretation: odds

A more natural way of interpreting the coefficients of a logistic regression is based on **odds**

Using basic algebra, the model can also be written as:

$$\underbrace{\frac{p_i}{1 - p_i}}_{\text{odds}} = e^{\beta_0 + \beta_1 \text{ fare}_i} = e^{\beta_0} e^{\beta_1 \text{ fare}_i} \qquad \text{[odds]}$$

> We need to exponentiate the coefficients to interprete them as changes in the **odds**. Find more details in the math appendix

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Exponentating the coefficients

```r
#Coefficients for the odds
tidy(model_titanic_logistic, exponentiate = TRUE) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

$e^{\hat{\beta}_0}$

$e^{\hat{\beta}_1}$

term	estimate	std.error	statistic	p.value
(Intercept)	0.390	0.095	-9.895	0
fare	1.015	0.002	6.810	0

Slope: $e^{\hat{\beta}_1} = e^{0.0152} = 1.015$: an increase of 1 dollar in the `fare` paid is associated with an increase of the odds of surviving by a factor of 1.015 or an increase of 1.5% percent

Case 2: one categorical variable

$$\log\left(\frac{P(\text{survived}_i = 1 |\ X_{\text{male,i}})}{1 - P(\text{survived}_i = 1 |\ X_{\text{male,i}})}\right) = \beta_0 + \beta_1 X_{\text{male,i}}$$

where $X_{\text{male,i}} = 1$ if passenger i is registered as a male and $X_{\text{male,i}} = 0$ if registered as female.

Again, 2 logistic regressions in 1!!

$$\text{logit}(p_{\text{survival}}) = \beta_0 + \beta_1 \times X_{\text{male}} = \begin{cases} \beta_0 & \text{if } X_{\text{male}} = 0 \\ \beta_0 + \beta_1 & \text{if } X_{\text{male}} = 1 \end{cases}$$

Therefore, for the odds of surviving:

$$\frac{p_{\text{survival}}}{1 - p_{\text{survival}}} = \begin{cases} e^{\beta_0} & \text{if } X_{\text{male}} = 0 \\ e^{\beta_0 + \beta_1} & \text{if } X_{\text{male}} = 1 \end{cases}$$

> find more details in the math appendix

in R

Scroll down to see full content

```r
model_titanic_logistic_sex <- glm(survived ~ sex, data = titan, family = bi

tidy(model_titanic_logistic_sex) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	1.057	0.129	8.191	0
sexmale	-2.514	0.167	-15.036	0

```r
tidy(model_titanic_logistic_sex, exponentiate = TRUE) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```


term	estimate	std.error	statistic	p.value
(Intercept)	2.877	0.129	8.191	0
sexmale	0.081	0.167	-15.036	0

- Reference level: female, note the dummy variable name sexmale

Interpretation (in terms of odds)

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- **Intercept:** $e^{\hat{\beta}_0} = e^{1.057} = 2.8765$, a female passenger had a 2.8765 odds of surviving (i.e., the proportion of survivals relative to the proportion of deaths in the sample)

- **Slope**: $e^{\hat{\beta}_1} = e^{-2.514} = 0.08097$, the odds of surviving for a male passenger are **multiplied** by a factor of 0.08097; a decrease of 91.90% (note $(0.08097 - 1) \times 100 = -91.90$)

I

Odds ratio: note that $e^{\hat{\beta}_1} = 0.08097 = \frac{\text{odds}_{\text{male}}}{\text{odds}_{\text{female}}}$

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- turning to **odds of failure**: the male passangers' **odds** of **dying** were $e^{2.514} = 12.3542$ **times** higher than of those of females

More details about these equivalences are given in the math appendix

Case 3: one categorical and one numerical covariate

Let's fit a logistic regression using `sex` and `fare` (omiting subscript i for simplicity)

The additive model

$$\log\left(\frac{P(\text{surv} = 1 |\ \text{fare}, X_{\text{male}})}{1 - P(\text{surv} = 1 |\ \text{fare}, X_{\text{male}})}\right) = \beta_0 + \beta_1 \underline{X_{\text{male}}} + \beta_2 \times \underline{\text{fare}}$$

$\underbrace{\log\ \text{odds}}$ $(E(y_i))$

$p(y_i = 1)$

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Again, 2 logistic regressions in 1 with **equal** slope!

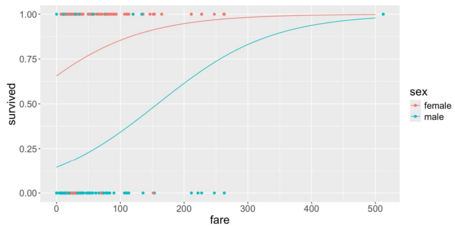
$$\text{logit}(p_{\text{survival}}) = \beta_0 + \beta_1 \times X_{\text{male}} + \beta_2 \times \text{fare}$$

$$\log\left(\frac{p_s}{1 - p_s}\right) = \begin{cases} \beta_0 + \beta_2 \times \text{fare}, & \text{if } X_{\text{male}} = 0 \\ \beta_0 + \beta_1 + \beta_2 \times \text{fare}, & \text{if } X_{\text{male}} = 1 \end{cases}$$

> same *slope* (β_2), different *intercepts* (β_0 vs ($\beta_0 + \beta_1$))!

For the odds of surviving:

$$\frac{p_{\text{survival}}}{1 - p_{\text{survival}}} = \begin{cases} e^{\beta_0} e^{\beta_2 \text{fare}}, & \text{if } X_{\text{male}} = 0 \\ e^{\beta_0 + \beta_1} e^{\beta_2 \text{fare}}, & \text{if } X_{\text{male}} = 1 \end{cases}$$



The curves won't look "parallel", even if the linear components have the same slopes

$$p_i = \begin{cases} \frac{e^{\beta_0+\beta_2\times \text{fare}_i}}{1+e^{\beta_0+\beta_2\times \text{fare}_i}} & \text{if } X_{\text{male, i}} = 0 \\ \frac{e^{\beta_0+\beta_1+\beta_2\times \text{fare}_i}}{1+e^{\beta_0+\beta_1+\beta_2\times \text{fare}_i}} & \text{if } X_{\text{male, i}} = 1 \end{cases}$$

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in R

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```r
model_titanic_logistic_additive <- glm(survived ~ sex + fare, data = titan,
                                        family = binomial)

tidy(model_titanic_logistic_additive) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	0.647	0.149	4.358	0
sexmale	-2.423	0.171	-14.208	0
fare	0.011	0.002	4.886	0

$\log(odds) = \beta_0 + \beta_1 fare + \beta_2 sex$

additive

```r
tidy(model_titanic_logistic_additive, exponentiate = TRUE) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	1.910	0.149	4.358	0
sexmale	0.089	0.171	-14.208	0
fare	1.011	0.002	4.886	0

$odds = e^{\beta_0} * e^{\beta_1 fare} * e^{\beta_2 sex}$

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Interpretation (for odds ratios)

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Keeping the `fare` value constant, **at any value**:

- the men's expected **odds of surviving** are $e^{-2.42276} = 0.0887$ **times** that of a female

- equivalently, male passengers have only 8.87% of the odds of surviving a female passenger.

  - or in other words, the men's expected **odds of dying** are $e^{2.42276} = 11.2769$ **times** that of a female

> note that we can "keep `fare` constant" since this is an **additive** model. More details in the math appendix

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Interpretation (cont.)

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Keeping the `sex` constant, **for either male or female**:

- increasing the fare in 1 dollar is associated with an increase of the odds of surviving **by a factor of** 1.0113.
- increasing the fare in 1 dollar is associated with an increase of the odds of surviving of $(1.0113 - 1) \times 100\% = 1.13\%$.

Case 4: one categorical and one numerical covariate

Model with interaction

$$\log\left(\frac{p_s}{1-p_s}\right) = \beta_0 + \beta_1 X_{\text{male}} + \beta_2 \times \text{fare} + \beta_3 \times X_{\text{male}} \times \text{fare}$$

Again, 2 logistic regressions in 1 with **different** slopes!

$$\text{logit}(p_s) = \beta_0 + \beta_1 \times X_{\text{male}} + \beta_2 \times \text{fare} + \beta_3 \times X_{\text{male}} \times \text{fare}$$

$$\log\left(\frac{p_s}{1-p_s}\right) = \begin{cases} \beta_0 + \beta_2 \times \text{fare}, & \text{if } X_{\text{male}} = 0 \\ (\beta_0 + \beta_1) + (\beta_2 + \beta_3) \times \text{fare}, & \text{if } X_{\text{male}} = 1 \end{cases}$$

> different *slope* (β_2 vs ($\beta_2 + \beta_3$)) , different *intercepts* (β_0 vs ($\beta_0 + \beta_1$))!

For the odds of surviving:

$$\frac{p_{\text{survival}}}{1 - p_{\text{survival}}} = \begin{cases} e^{\beta_0} e^{\beta_2 \text{fare}}, & \text{if } X_{\text{male}} = 0 \\ e^{(\beta_0+\beta_1)} e^{(\beta_2+\beta_3)\text{fare}}, & \text{if } X_{\text{male}} = 1 \end{cases}$$

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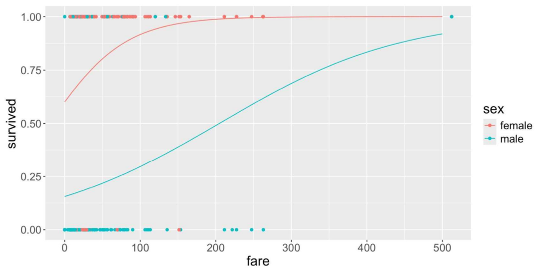

$$p_i = \begin{cases} \frac{e^{\beta_0+\beta_2\times \text{fare}_i}}{1+e^{\beta_0+\beta_2\times \text{fare}_i}} & \text{if } X_{\text{male, i}} = 0 \\ \frac{e^{(\beta_0+\beta_1)+(\beta_2+\beta_3)\times \text{fare}_i}}{1+e^{(\beta_0+\beta_1)+(\beta_2+\beta_3)\times \text{fare}_i}} & \text{if } X_{\text{male, i}} = 1 \end{cases}$$

in R

Scroll down to see full content

```r
model_titanic_logistic_int <- glm(survived ~ sex * fare, data = titan,
                                  family = binomial)

tidy(model_titanic_logistic_int) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	0.408	0.190	2.150	0.032
sexmale	-2.099	0.230	-9.116	0.000
fare	0.020	0.005	3.701	0.000
sexmale:fare	-0.012	0.006	-1.958	0.050

significant interaction

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```r
tidy(model_titanic_logistic_int, exponentiate = TRUE) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	1.504	0.190	2.150	0.032
sexmale	0.123	0.230	-9.116	0.000
fare	1.020	0.005	3.701	0.000

Interpretation (for odds ratios)

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In a **model with interactions**, the expected change in one variable **depends on** the value or levels the other variable

We **CAN'T** say "keeping the `fare` value constant" or "for either male or female"

- *for female passengers*, increasing the fare in 1 dollar is associated with an increase of the odds of surviving **by a factor of** 1.020.

- *for female passengers*, increasing the fare in 1 dollar is associated with an increase of the odds of surviving of 2%.

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Interaction term

The interpretation of the interaction terms becomes a bit complicated.

Instead, you can calculate the two estimated curves and interprete them accordingly.

From the `tidy` table (with `exponentiate = FALSE` or default), we get

$$\hat{\beta}_0 = 0.408,\ \hat{\beta}_1 = -2.099,\ \hat{\beta}_2 = 0.020,\ \hat{\beta}_3 = -0.012$$

Then,

$$\begin{aligned} \hat{\beta}_0 &= 0.408 \\ \hat{\beta}_2 &= 0.020 \\ \hat{\beta}_0 + \hat{\beta}_1 &= 0.408 - 2.099 = -1.691 \\ \hat{\beta}_2 + \hat{\beta}_3 &= 0.020 - 0.012 = 0.008 \end{aligned}$$

$$\frac{p_i}{1-p_i} = \begin{cases} e^{0.408+0.020\times \text{ fare}_i} & \text{if } X_{\text{male, i}} = 0 \\ e^{-1.691+0.008\times \text{ fare}_i} & \text{if } X_{\text{male, i}} = 1 \end{cases}$$

- *for male passengers*, increasing the fare in 1 dollar is associated with an increase of the odds of surviving **by a factor of** $e^{0.008} = 1.008$.

- *for male passengers*, increasing the fare in 1 dollar is associated with an increase in the odds of surviving of 0.8%.

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Using exponentiated coefficients

Note that you can also recover both curves using the exponentiated coefficients, from the `tidy` table, with `exponentiate = TRUE`.

$$e^{\hat{\beta}_0} = 1.504, \; e^{\hat{\beta}_1} = 0.123, \; e^{\hat{\beta}_2} = 1.020, \; e^{\hat{\beta}_3} = 0.988$$

We need to multiply (instead of summing):

$$\begin{aligned} e^{\hat{\beta}_0+\hat{\beta}_1} &= e^{-1.691} = 0.184 \\ e^{\hat{\beta}_0} \; e^{\hat{\beta}_1} &= 1.504 \times 0.123 = 0.185 \\ e^{\hat{\beta}_2+\hat{\beta}_3} &= e^{0.008} = 1.008 \\ e^{\hat{\beta}_2} \;\; e^{\hat{\beta}_3} &= 1.020 \times 0.988 = 1.008 \end{aligned}$$

> which are equivalent, up to rounding errors

Inference

Inference for GLMs, and logistic regression in particular, are based on theorectical results from the CLT.

> if the sample size is large, the *Normal* distribution is used to obtain boudaries of confidence intervals and p-values

$H_0: \beta_j = 0$ vs $H_1: \beta_j \neq 0$

Under H_0: $$\frac{\hat{\beta}_j - 0}{\hat{\text{SE}}(\hat{\beta}_j)} \sim N(0, 1)$$ (approximately)

> This is called the **Wald's statistic**

test statistics

95% C.I. for β_j: $\hat{\beta}_j \pm 1.96 \cdot SE(\hat{\beta}_j)$

statistical thinking

① CLT: If $X_1, X_2, \cdots, X_n$ i.i.d., then

~~$(\bar{X} - \mu) \rightarrow$~~

$\bar{X} \xrightarrow{d} N(\mu, \frac{\sigma^2}{n})$

$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$ $n \to \infty$

② MLE $\hat{\beta}$.

$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, I^{-1}(\beta))$ $n \to \infty$

$\Longleftrightarrow \frac{\hat{\beta} - \beta}{SE(\hat{\beta})} \xrightarrow{d} N(0, 1)$

$p = 0.049$?

The confidence intervals of $(1 - \alpha)\%$ for the coefficients of the log-odds model (canonical link) are

$$CI\left(\hat{\beta}_j, 1 - \alpha\right) = \hat{\beta}_j \pm z_{\alpha/2} \hat{\text{SE}}(\hat{\beta}_j)$$

You can also use the Wald's statistic to test $H_0 : \hat{\beta}_j = 0$ vs $H_1 : \hat{\beta}_j \neq 0$

But again, R will do all this for you!

in R

```
tidy(model_titanic_logistic_additive,
    conf.int = TRUE, conf.level = 0.9) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.647	0.149	4.358	0	0.405	0.894
sexmale	-2.423	0.171	-14.208	0	-2.707	-2.145
fare	0.011	0.002	4.886	0	0.008	0.015

For `fare`: $p < 0.001$ indicating significant evidence to **reject the null hypothesis** that states that there is no association between `fare` and the log-odds of surviving, for either female or male passengers.

Inference for odds of success

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You can also get confidence intervals for the exponenciated coefficients

```
tidy(model_titanic_logistic_additive,
    conf.int = TRUE, conf.level = 0.9, exponentiate = TRUE)%>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.910	0.149	4.358	0	1.499	2.444
sexmale	0.089	0.171	-14.208	0	0.067	0.117
fare	1.011	0.002	4.886	0	1.008	1.015

We can be 90% confident that an increase of 1 dollar in `fare` is associated with a 0.8% to 1.5% increase in the odds of surviving for either male or female passengers.

> since $(1.008 - 1) \times 100\% = 0.8\%$ and $(1.015 - 1) \times 100\% = 1.5\%$

Summary

$$\frac{E(y|x)}{E(y)}$$

- The (conditional) expectation of a binary response is the (conditional) *probability of success*.

- A LR can not be used to model the conditional expectation of a binary response since its range extends beyond the interval $(0, 1)$

- Instead, one can model a function of the conditional probability. A common choice in logistic regression is to use the *logit* function (logarithm of odds)

$\rightarrow f(x) = \log \frac{x}{1-x}$

Summary (cont.)

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The interpretation of the coefficients depends on the type of variables and the form of the model:

The raw coefficients (of the log-odds model) are interpreted as:

- log-odds of a reference group
- difference of log-odds of a treatment vs a control group
- changes in log-odds per unit change in the input

$\log(\cancel{odds}\,odds) = \beta_0 + \beta_1 X$

The exponentiated coefficients (of the odds model) are interpreted as:

- odds of a reference group
- odds ratio of a treatment vs a control group
- multiplicative changes in odds per unit change in the input

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$odds = e^{\beta_0} \cdot e^{\beta_1 X}$

Summary (cont.)

- The estimated logistic model can be used to make inference using the Wald's satistic: confidence intervals and tests

0 1

- In the next lecture we'll look at
  - fitted values
  - residuals
  - diagnostics

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